Fiscal Year:	FY 2019	**Task Last Updated:**	FY 02/06/2019
PI Name:	Ganguli, Gurudas Ph.D.		
Project Title:	Understanding the Frequency Synchronization Physics in PK-4 Experiment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Washington	**State:**	DC
Zip Code:	20375-0001	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	04/15/2017	**End Date:**	04/15/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Crabtree, Chris Ph.D. (Naval Research Laboratory)		
Grant/Contract No.:	JPL 1573108		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this proposal is to provide theoretical support for a separate PK-4 (Plasma Krystall-4) experimental project, “Nonlinear Waves in Dusty Plasmas.” The experimental effort, led by Principal Investigator (PI) Goree, is an investigation of nonlinear synchronization, with the advance that it will study not point oscillators, but waves that propagate a distance of many wavelengths. The literature is lacking theory suitable for comparison to the experiment, perhaps because the experiments at this point are still very new. In particular, theory is needed for the case, beyond the point oscillator description that is most common, for synchronization in the realm of propagating waves. Beyond just supporting the PK-4 experiment, the theoretical advances we propose can have a broad interdisciplinary impact. Since synchronization, frequency entrainment, and frequency pulling are nonlinear phenomena that impact a variety of physical, chemical, and biological systems, the proposed investigation will be relevant to a broader area of contemporary interest and enhance the science foundation for nonlinear phenomena in plasma physics. A plasma system normally constitutes a highly complex medium that can display a rich variety of nonlinear phenomena arising out of a host of intrinsic oscillations and other non-stationary states. A clear identification of phenomena like synchronization or frequency entrainment in such a medium often poses serious experimental challenges. Many past studies on such systems have therefore relied on driving a single plasma system with an external frequency source. There have been very few studies devoted to the investigation of mutual synchronization between two plasma systems except where this phenomenon was studied between two double plasma devices and where it was investigated in two electrical discharges. There is no theoretical framework to account for these phenomena in a realistic dispersive plasma system that includes natural growth and dissipation of waves. Therefore, the primary focus of this proposal is to develop such a framework and apply it to understand the dedicated PK-4 dusty plasma experiment on frequency synchronization. The validated model will then be useful for general application to other experiments. Some of the science questions that we will address in our proposed theoretical investigation are the following: • What are the parametric domains in the state space of the external driver frequency and amplitude that lead to frequency entrainment, frequency pulling, and synchronization of the dust acoustic waves in the PK-4 plasma? • How do dispersion and nonlinearity of the dusty plasma medium influence the nature of the synchronization process? • Can the non-harmonic synchronized state, as observed in ground-based experiments, be predicted by a continuum model of the dusty plasma's constituents? • What determines the threshold amplitude for synchronization, and how is the threshold affected by dust-dust correlations (i.e., the strength of the Coulomb parameter)? • How does viscous dissipation in the medium affect synchronization and influence the nature of the synchronized excitation? • Can one observe phase-flip transitions in the synchronization, i.e., a sudden transition between in-phase synchronization to pi out-of-phase synchronization? • How does a broad-spectrum driver (e.g., signals from another plasma source) alter the synchronization scenario?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Beyond just supporting the PK-4 experiment, the theoretical advances in this project will have a broad interdisciplinary impact. Since synchronization, frequency entrainment, and frequency pulling are nonlinear phenomena that impact a variety of physical, chemical, and biological systems, the investigation is relevant to a broader area of contemporary interest.
Task Progress:	We are in the second year of the program. We completed a basic review of the related materials and understood the connection with the PK-4 experiment. We interacted with the experimental team of Professor Goree through a visit to his laboratory in the University of Iowa and at other venues. We applied our nonlinear model of the dust acoustic waves constructed in the previous year to existing ground experiments [1,2], which did not include external modulations. We found good fit of the experimental data with an exact nonlinear solution of the KdV equation, namely the cnoidal wave solution [3], and obtained good agreement between the experimental observations and the model solution over a large range of the background pressure. While the model reproduces the profile of the observed nonlinear wave quite well there is some discrepancy with the magnitude of the frequency. This may be due to the fact that the values of the experimental parameters used in the model may correspond to the central region of the plasma discharge where it is easier to make measurements whereas the dust cloud exists in the sheath region where it is difficult to make accurate measurements. A manuscript detailing the results has been published in the Physics of Plasmas [4] [Ed. note: see also Bibliography section]. This publication was selected as the Editor’s Choice by the journal. In addition we presented the results of this investigation in the 34th Annual Meeting of the American Society for Gravitational and Space Research in October 2018 in Washington, DC. References: [1] T. M. Flanagan and J. Goree, Observation of the spatial growth of self-excited dust-density waves, Physics of Plasmas, 17 (2010) 123702. [2] T. M. Flanagan and J. Goree, Development of nonlinearity in a growing self- excited dust-density wave, Physics of Plasmas, 18 (2011) 013705. [3] Pava et al , Advances in Differential Equations 11 (2006) 13211374 [4] Liu, B., J. Goree, T.M. Flanagan, A. Sen, S.K. Tiwari, G. Ganguli, and C. Crabtree, Experimental observation of cnoidal waveform of nonlinear dust acoustic waves, Phys Plasmas, 25, 113701, 2018: doi.org/10.1063/1.5046402
Bibliography Type:	Description: (Last Updated: 03/22/2023)
Articles in Peer-reviewed Journals	Liu B, Goree J, Flanagan TM, Sen A, Tiwari SK, Ganguli G, Crabtree C. "Experimental observation of cnoidal waveform of nonlinear dust acoustic waves." Physics of Plasmas. 2018 Nov;25(11):113701. https://doi.org/10.1063/1.5046402 , Nov-2018

Awards	Liu B, Goree J, Flanagan TM, Sen A, Tiwari SK, Ganguli G, Crabtree C. "Editor's Choice, November 1, 2018 Physics of Plasma journal for article, 'Experimental observation of cnoidal waveform of nonlinear dust acoustic waves.' Physics of Plasmas. 2018 Nov;25(11):113701. https://doi.org/10.1063/1.5046402 " Nov-2018